

## Phase 8:Project Demonstration

### SmartLenderAI-Powered Credit Card Approval System

| Role             | Name                       |
|------------------|----------------------------|
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#### Demo Scenario 1 – Automated Screening

A credit analyst enters an applicant's financial profile into the web application. The application returns an instant approve/reject prediction, letting the analyst prioritize which applications need closer review.

#### Demo Scenario 2 – High-Risk Identification

A compliance officer screens applicants with past-due loan records. The feature engineering pipeline converts multi-class payment status into a binary risk score, so high-risk applicants are clearly flagged.

#### Demo Scenario 3 – Self-Service Eligibility Check

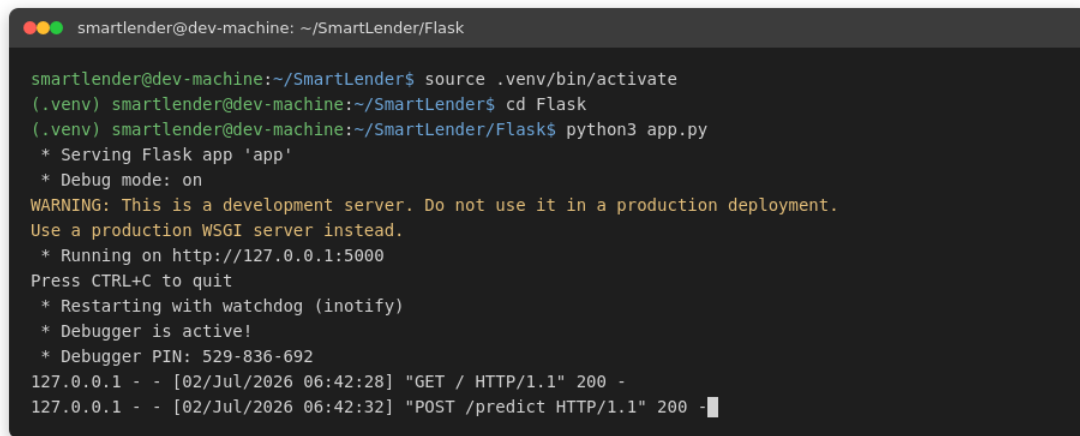
A prospective customer enters their own details before formally applying and receives an instant likelihood of approval, reducing the number of applications that would otherwise be rejected outright.

#### Conclusion & Future Scope

SmartLender shows that a tuned XGBoost model can automate a meaningful share of the credit approval workload. Future work includes adding explainability for each decision and monitoring the model for drift as new applicant data comes in.

## Terminal Execution

The screenshot below shows the Flask development server starting successfully and serving requests on local port 5000.

A terminal window with a dark background and light text. The prompt is 'smartlender@dev-machine: ~/SmartLender/Flask'. The user enters 'source .venv/bin/activate', then 'cd Flask', and finally 'python3 app.py'. The output shows the server starting, including a warning about production use, the port (5000), and a debugger PIN. Two HTTP requests are shown: a GET request to '/' and a POST request to '/predict', both returning 200.

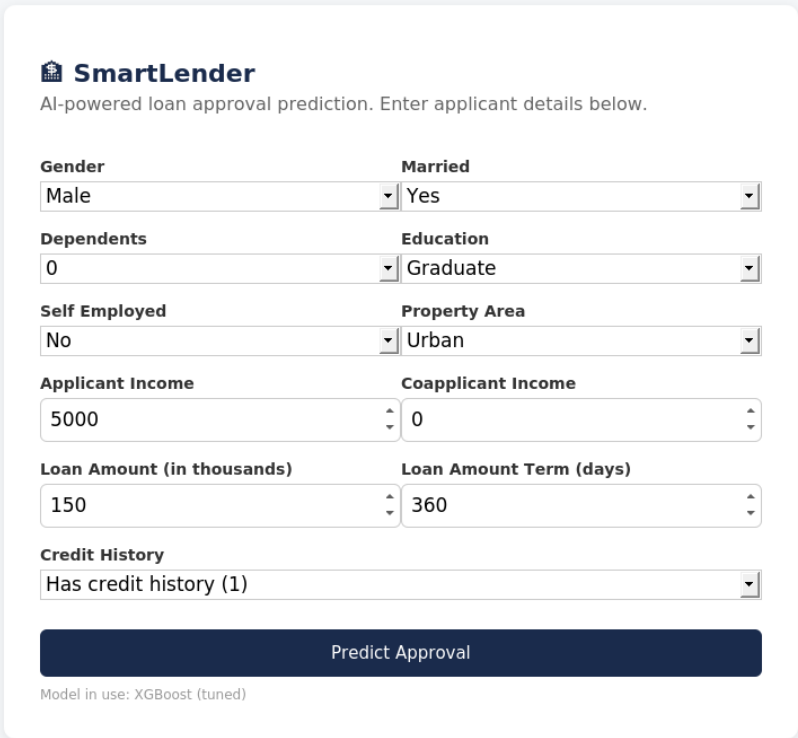
```
smartlender@dev-machine: ~/SmartLender/Flask

smartlender@dev-machine:~/SmartLender$ source .venv/bin/activate
(.venv) smartlender@dev-machine:~/SmartLender$ cd Flask
(.venv) smartlender@dev-machine:~/SmartLender/Flask$ python3 app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (inotify)
* Debugger is active!
* Debugger PIN: 529-836-692
127.0.0.1 - - [02/Jul/2026 06:42:28] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Jul/2026 06:42:32] "POST /predict HTTP/1.1" 200 -
```

*Figure 5: Flask server running in terminal*

## Web Interface – Input Form

The applicant enters demographic and financial details through the web form shown below.



**SmartLender**  
AI-powered loan approval prediction. Enter applicant details below.

|                                   |                                |
|-----------------------------------|--------------------------------|
| <b>Gender</b>                     | <b>Married</b>                 |
| Male                              | Yes                            |
| <b>Dependents</b>                 | <b>Education</b>               |
| 0                                 | Graduate                       |
| <b>Self Employed</b>              | <b>Property Area</b>           |
| No                                | Urban                          |
| <b>Applicant Income</b>           | <b>Coapplicant Income</b>      |
| 5000                              | 0                              |
| <b>Loan Amount (in thousands)</b> | <b>Loan Amount Term (days)</b> |
| 150                               | 360                            |
| <b>Credit History</b>             |                                |
| Has credit history (1)            |                                |

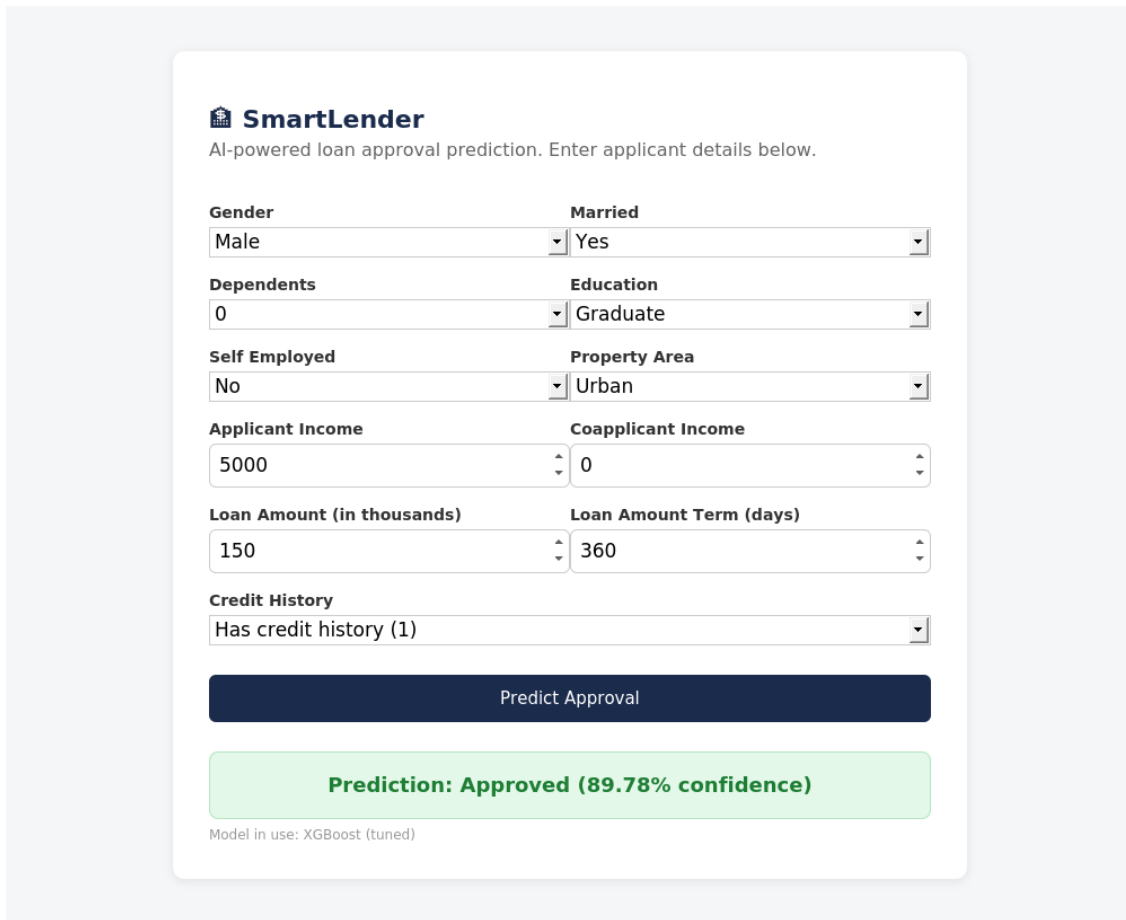
Predict Approval

Model in use: XGBoost (tuned)

Figure 6: Input form interface

## Web Interface – Prediction Output

After submission, the system returns the approve/reject decision along with the tuned XGBoost model's confidence percentage.



The image shows a web interface for 'SmartLender', an AI-powered loan approval prediction system. The interface is a light gray card with a white background. At the top, it has the 'SmartLender' logo and the text 'AI-powered loan approval prediction. Enter applicant details below.' Below this, there are several input fields arranged in a grid. The first row has 'Gender' (Male) and 'Married' (Yes). The second row has 'Dependents' (0) and 'Education' (Graduate). The third row has 'Self Employed' (No) and 'Property Area' (Urban). The fourth row has 'Applicant Income' (5000) and 'Coapplicant Income' (0). The fifth row has 'Loan Amount (in thousands)' (150) and 'Loan Amount Term (days)' (360). The sixth row has 'Credit History' (Has credit history (1)). Below these fields is a dark blue button labeled 'Predict Approval'. Below the button is a green box with the text 'Prediction: Approved (89.78% confidence)'. At the bottom, there is a small text 'Model in use: XGBoost (tuned)'.

**SmartLender**  
AI-powered loan approval prediction. Enter applicant details below.

|                                   |                                |
|-----------------------------------|--------------------------------|
| <b>Gender</b>                     | <b>Married</b>                 |
| Male                              | Yes                            |
| <b>Dependents</b>                 | <b>Education</b>               |
| 0                                 | Graduate                       |
| <b>Self Employed</b>              | <b>Property Area</b>           |
| No                                | Urban                          |
| <b>Applicant Income</b>           | <b>Coapplicant Income</b>      |
| 5000                              | 0                              |
| <b>Loan Amount (in thousands)</b> | <b>Loan Amount Term (days)</b> |
| 150                               | 360                            |
| <b>Credit History</b>             |                                |
| Has credit history (1)            |                                |

Predict Approval

**Prediction: Approved (89.78% confidence)**

Model in use: XGBoost (tuned)

Figure 7: Prediction output display